MATHS BRIDGE ACADEMY

PRELIM 1 – MATHEMATICS STANDARD

MARKING SCHEME

CLASS X (CBSE) | Max. Marks: 80

SECTION: A (Solution of MCQs of 1 Mark each, no part marks)

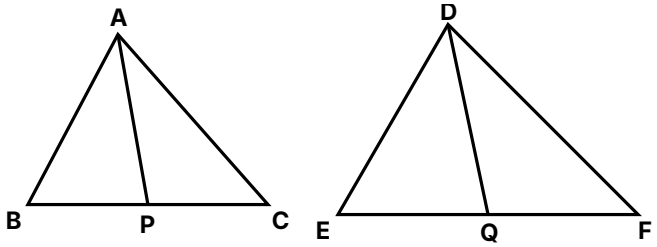
Q. No.	Ans.	Hints/Solution	Marks
1.	**(C) 3**	LCM (a, b, c) = $2^2 \times 3^x \times 5 \times 7 = 3780$ $\Rightarrow 3^x = 27 \Rightarrow x = 3$ *[LCM to be read as 3780; printed value 37800 gives no valid x]*	**1**
2.	**(A) 2**	Distance of a point from y-axis = |x-coordinate| = 2	**1**
3.	**(B) $\neq \frac{15}{4}$**	For lines not to be parallel: $\frac{3}{2} \neq \frac{2k}{5}$ $\Rightarrow k \neq \frac{15}{4}$	**1**
4.	**(C) 6 cm**	AB + CD = BC + AD $\Rightarrow$ AB + 4 = 7 + 3 $\Rightarrow$ AB = 6 cm	**1**
5.	**(D) $\frac{1}{x}$**	$\sec^2\theta - \tan^2\theta = 1$ $\Rightarrow (\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = 1$ $\Rightarrow \sec\theta - \tan\theta = \frac{1}{x}$	**1**
6.	**(D)**	$(x + 2)(x + 1) = x^2 + 2x + 3$ $\Rightarrow x^2 + 3x + 2 = x^2 + 2x + 3$ $\Rightarrow x - 1 = 0$, which is not a quadratic equation.	**1**
7.	**(D) $8[\frac{\pi}{6} - \frac{\sqrt{3}}{4}]$ cm^2**	Chord = radius = 1 cm $\Rightarrow$ angle subtended at centre = 60° Area of one segment = $\frac{60}{360} \times \pi \times 1^2 - \frac{\sqrt{3}}{4} \times 1^2 = \frac{\pi}{6} - \frac{\sqrt{3}}{4}$ 4 dotted regions = 8 segments $\therefore$ Total area = $8[\frac{\pi}{6} - \frac{\sqrt{3}}{4}]$ cm^2	**1**
8.	**(B) $\frac{31}{36}$**	Total outcomes = 36 Sum 8: (2, 6), (3, 5), (4, 4), (5, 3), (6, 2) i.e. 5 outcomes P(not getting sum 8) = $1 - \frac{5}{36} = \frac{31}{36}$	**1**
9.	**(B) 12°**	$\sin 5x = \frac{\sqrt{3}}{2} = \sin 60°$ $\Rightarrow 5x = 60° \Rightarrow x = 12°$	**1**

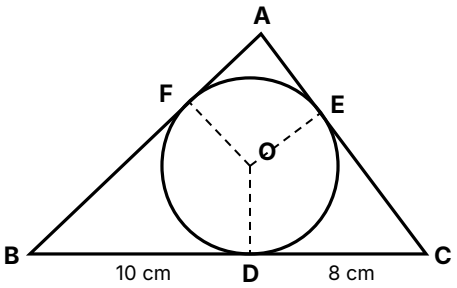
10.	**(C) 4**	Let numbers be 81a and 81b, where a, b are co-prime. $81(a + b) = 1215 \Rightarrow a + b = 15$ Co-prime pairs: (1, 14), (2, 13), (4, 11), (7, 8) $\therefore$ Number of pairs = 4	**1**
11.	**(D) 5 cm**	$\frac{1}{3} \times 51 \times h = 85$ $\Rightarrow h = 5$ cm	**1**
12.	**(D)**	For equal zeroes, $b^2 - 4ac = 0 \Rightarrow b^2 = 4ac$ As $b^2 > 0$, $ac > 0$ $\therefore$ c and a must have same signs.	**1**
13.	**(C) 231**	Area of sector $= \frac{1}{2} \times l \times r = \frac{1}{2} \times 22 \times 21 = 231\ cm^2$	**1**
14.	**(C) 18 cm**	Perimeter of $\triangle DEF = 9 + 6 + 12 = 27$ cm $\frac{\text{Perimeter of } \triangle ABC}{\text{Perimeter of } \triangle DEF} = \frac{AB}{DE} = \frac{6}{9}$ $\Rightarrow$ Perimeter of $\triangle ABC = \frac{2}{3} \times 27 = 18$ cm	**1**
15.	**(B) $\frac{9}{4}$**	Total letters = 11, vowels (a, e, a, i) = 4 P(vowel) $= \frac{4}{11} = \frac{2}{2x + 1}$ $\Rightarrow 4x + 2 = 11 \Rightarrow x = \frac{9}{4}$	**1**
16.	**(C) Parallelogram**	$AB = CD = 6$, $BC = DA = \sqrt{360}$ Diagonals: $AC = 18$, $BD = \sqrt{468}$ (not equal) Opposite sides equal, diagonals unequal $\Rightarrow$ Parallelogram	**1**
17.	**(A)**	If each observation is increased by 2, the median is also increased by 2. New median = 20.5 + 2 = 22.5	**1**
18.	**(A) 40 cm**	Length of tangent $= \sqrt{41^2 - 9^2} = \sqrt{1600} = 40$ cm	**1**
19.	**(A)**	Both **Assertion (A)** and **Reason (R)** are true and **Reason (R)** is the correct explanation of **Assertion (A)**.	**1**
20.	**(A)**	$\cos A = 1 - \cos^2 A = \sin^2 A \Rightarrow \sin^2 A + \sin^4 A = \cos A + \cos^2 A = 1$ Both **Assertion (A)** and **Reason (R)** are true and **Reason (R)** is the correct explanation of **Assertion (A)**.	**1**

Section – B

[This section comprises of solution of very short answer type questions (VSA) of 2 marks each]

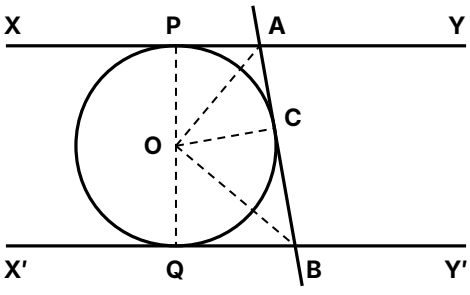
Q. No.	HINTS/SOLUTION	Marks
21. (A)	Here a = 8, d = 2, n = 60	

	$a_{51} = 8 + 50 \times 2 = 108$	½
	$a_{60} = 8 + 59 \times 2 = 126$	½
	Sum of last 10 terms $= \frac{10}{2}[108 + 126]$	½
	$= 1170$	½
	OR	
21. (B)	Here $a = 6, d = 7, a_n = 230$	
	$6 + (n - 1) \times 7 = 230 \Rightarrow n = 33$	**1**
	Middle term $= \frac{33 + 1}{2} = 17^{th}$ term	½
	$\therefore a_{17} = 6 + 16 \times 7 = 118$	½
22.	$\sin(A + B) = 1 = \sin 90° \Rightarrow A + B = 90°$ --- (i)	½
	$\cos(A - B) = \frac{\sqrt{3}}{2} = \cos 30° \Rightarrow A - B = 30°$ --- (ii)	½
	Solving (i) and (ii), we get	
	$A = 60°$	½
	$B = 30°$	½
23.		
	$\triangle ABC \sim \triangle DEF \Rightarrow \frac{AB}{DE} = \frac{BC}{EF}$ and $\angle B = \angle E$	½
	AP and DQ are medians $\Rightarrow BC = 2BP, EF = 2EQ$ $\therefore \frac{AB}{DE} = \frac{2BP}{2EQ} = \frac{BP}{EQ}$	½
	In $\triangle ABP$ and $\triangle DEQ$, $\frac{AB}{DE} = \frac{BP}{EQ}$ and $\angle B = \angle E$	
	$\therefore \triangle ABP \sim \triangle DEQ$ (SAS criterion)	½
	$\Rightarrow \frac{AB}{DE} = \frac{AP}{DQ}$	½
24. (A)	Let $\angle A = \theta_1, \angle B = \theta_2, \angle C = \theta_3$	
	$\theta_1 + \theta_2 + \theta_3 = 180°$ (angle sum property)	½
	Area grazed $= \frac{\theta_1 + \theta_2 + \theta_3}{360°} \times \pi r^2$	
	$= \frac{180°}{360°} \times \frac{22}{7} \times 14 \times 14$	**1**
	$= 308\ m^2$	½

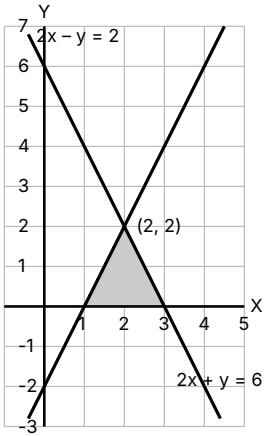
	OR	
24. (B)	Area of major sector = $\frac{270°}{360°} \times \pi \times 5^2 = \frac{75\pi}{4}$ cm^2	½
	Area of triangle = $\frac{1}{2} \times 5 \times 5 = \frac{25}{2}$ cm^2	½
	Area of major segment = Area of major sector + Area of triangle	½
	= $(\frac{75\pi}{4} + \frac{25}{2})$ cm^2	½
25.		
	BF = BD = 10 cm, CE = CD = 8 cm (tangents from an external point)	
	Let AE = AF = x cm	
	AB = (x + 10) cm, AC = (x + 8) cm, BC = 18 cm	½
	ar(△ABC) = ar(△OBC) + ar(△OCA) + ar(△OAB)	
	= $\frac{1}{2} \times 4 \times [18 + (x + 8) + (x + 10)] = 2(2x + 36)$	½
	2(2x + 36) = 90 ⇒ x = 4.5	½
	∴ AB = 14.5 cm and AC = 12.5 cm	½

Section – C

[This section comprises of solution of short answer type questions (SA) of 3 marks each]

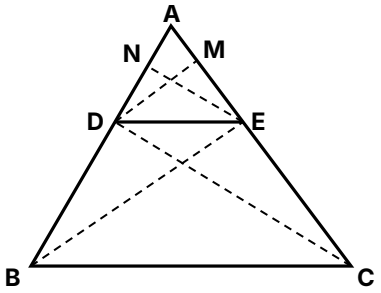
Q. No.	HINTS/SOLUTION	Marks
26.		
	Join OC.	
	In △OPA and △OCA,	
	OP = OC (radii)	
	AP = AC (tangents from external point A)	
	OA = OA (common)	
	∴ △OPA ≅ △OCA (SSS) ⇒ ∠POA = ∠COA --- (i)	**1**
	Similarly, △OQB ≅ △OCB ⇒ ∠QOB = ∠COB --- (ii)	**1**

	POQ is a diameter of the circle. $\therefore \angle POA + \angle COA + \angle COB + \angle QOB = 180°$ $\Rightarrow 2\angle COA + 2\angle COB = 180°$ [from (i) and (ii)] $\Rightarrow \angle COA + \angle COB = 90°$ $\Rightarrow \angle AOB = 90°$	½ ½
27.	Number of teachers in each room = HCF (36, 60, 84) $36 = 2^2 \times 3^2$ $60 = 2^2 \times 3 \times 5$ $84 = 2^2 \times 3 \times 7$ $\therefore$ HCF (36, 60, 84) $= 2^2 \times 3 = 12$ Minimum number of rooms $= \frac{36}{12} + \frac{60}{12} + \frac{84}{12} = 3 + 5 + 7$ $= 15$	½ **1** ½ ½ ½
28.	$2x^2 - (1 + 2\sqrt{2})x + \sqrt{2}$ $= 2x^2 - x - 2\sqrt{2}x + \sqrt{2}$ $= x(2x - 1) - \sqrt{2}(2x - 1)$ $= (2x - 1)(x - \sqrt{2})$ $\therefore$ Zeroes are $\frac{1}{2}$ and $\sqrt{2}$ Sum of zeroes $= \frac{1}{2} + \sqrt{2} = \frac{1 + 2\sqrt{2}}{2} = \frac{-(\text{coefficient of } x)}{\text{coefficient of } x^2}$ Product of zeroes $= \frac{1}{2} \times \sqrt{2} = \frac{\sqrt{2}}{2} = \frac{\text{constant term}}{\text{coefficient of } x^2}$	 **1** **1** ½ ½
29. (A)	$\sin\theta + \cos\theta = \sqrt{2}$ Squaring both sides, we get $\sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta = 2$ $\Rightarrow \sin\theta\cos\theta = \frac{1}{2}$ Now, $\tan\theta + \cot\theta = \frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta} = \frac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta} = \frac{1}{\sin\theta\cos\theta}$ $= \frac{1}{\frac{1}{2}} = 2$	 **1** ½ **1** ½
	OR	
29. (B)	LHS $= \frac{\cos A - \sin A + 1}{\cos A + \sin A - 1}$ Dividing numerator and denominator by sinA, $= \frac{\cot A - 1 + \operatorname{cosec} A}{\cot A + 1 - \operatorname{cosec} A}$ $= \frac{(\cot A + \operatorname{cosec} A) - (\operatorname{cosec}^2 A - \cot^2 A)}{\cot A + 1 - \operatorname{cosec} A}$	 **1** ½

	$= \frac{(\text{cosec}A + \cot A)(1 - \text{cosec}A + \cot A)}{\cot A + 1 - \text{cosec}A}$	**1**
	$= \text{cosec}A + \cot A = RHS$	½
30.	Sample space = {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}	
	Total outcomes = 8	½
	Favourable outcomes for Vidhi (two heads in a row) = {HHH, HHT, THH}	½
	P(Vidhi drives) = $\frac{3}{8}$	½
	Favourable outcomes for Unnati (head immediately followed by tail) = {HHT, HTH, HTT, THT}	½
	P(Unnati drives) = $\frac{4}{8} = \frac{1}{2}$	½
	As $\frac{4}{8} > \frac{3}{8}$, Unnati has greater probability of driving the car.	½
31. (A)	Let monthly incomes of Aryan and Babban be ₹ 3x and ₹ 4x and their monthly expenditures be ₹ 5y and ₹ 7y respectively.	
	3x – 5y = 15000 --- (i)	½
	4x – 7y = 15000 --- (ii)	½
	Solving (i) and (ii), we get x = 30000 and y = 15000	**1**
	Monthly income of Aryan = ₹ 90,000	½
	Monthly income of Babban = ₹ 1,20,000	½
	OR	
31. (B)	2x + y = 6 **x** **0** **1** **3** y 6 4 0 2x – y – 2 = 0 **x** **0** **1** **2** y –2 0 2 	
	Correct table of values for both equations	**1**
	Correct graph	**1**
	Lines intersect at (2, 2). Solution: x = 2, y = 2	½
	Vertices of triangle formed with x-axis: (2, 2), (1, 0), (3, 0)	
	Area = $\frac{1}{2} \times 2 \times 2 = 2$ sq. units	½

Section – D

[This section comprises of solution of long answer type questions (LA) of 5 marks each]

Q. No.	HINTS/SOLUTION	Marks
32.	Let the original average speed of the train be x km/h.	
	Time taken to cover 63 km = $\frac{63}{x}$ h, time taken to cover 72 km = $\frac{72}{x+6}$ h	**1**
	$\frac{63}{x} + \frac{72}{x+6} = 3$	**1**
	$\Rightarrow 21(x+6) + 24x = x(x+6)$	
	$\Rightarrow x^2 - 39x - 126 = 0$	**1**
	$\Rightarrow (x-42)(x+3) = 0$	**1**
	$\Rightarrow x = 42, -3$	**½**
	Rejecting x = – 3, as speed cannot be negative. ∴ Original average speed of the train = 42 km/h	**½**
33.	Correct figure, given, to prove, construction	**1½**
		
	Given: In △ABC, DE ∥ BC. To prove: $\frac{AD}{DB} = \frac{AE}{EC}$	
	Construction: Join BE, CD. Draw EN ⊥ AB and DM ⊥ AC.	
	$\frac{ar(\triangle ADE)}{ar(\triangle BDE)} = \frac{\frac{1}{2} \times AD \times EN}{\frac{1}{2} \times DB \times EN} = \frac{AD}{DB}$ --- (i)	**½**
	$\frac{ar(\triangle ADE)}{ar(\triangle CED)} = \frac{\frac{1}{2} \times AE \times DM}{\frac{1}{2} \times EC \times DM} = \frac{AE}{EC}$ --- (ii)	**½**
	ar(△BDE) = ar(△CED) (same base DE and between same parallels) --- (iii)	
	From (i), (ii) and (iii), $\frac{AD}{DB} = \frac{AE}{EC}$	**½**
	In △PQR, LM ∥ QR	
	LQ = PQ – PL = 15.2 – 5.7 = 9.5 cm	**½**
	$\therefore \frac{PL}{LQ} = \frac{PM}{MR}$	**½**
	$\Rightarrow \frac{5.7}{9.5} = \frac{PM}{5.5}$	**½**

	⇒ PM = 3.3 cm	½
34. (A)	Slant height of cone (l) = $\sqrt{6^2 + 12^2} = \sqrt{180} = 6\sqrt{5}$ cm	**1**
	CSA of cone = $\pi r l = \pi \times 12 \times 6\sqrt{5} = 72\sqrt{5}\pi$ cm^2	**1**
	Area of remaining base = $\pi(12^2 - 4^2) = 128\pi$ cm^2	**1**
	CSA of cylindrical cavity + area of its top = $2\pi \times 4 \times 3 + \pi \times 4^2 = 40\pi$ cm^2	**1**
	Surface area of remaining solid = $72\sqrt{5}\pi + 128\pi + 40\pi$	½
	= $(168 + 72\sqrt{5})\pi$ cm^2	½
	OR	
34. (B)	Volume of cone = $\frac{1}{3} \times \pi \times 3^2 \times 12 = 36\pi$ cm^3	**1**
	Volume of ice-cream in the cone = $\frac{5}{6} \times 36\pi$	**1**
	= 30π cm^3	½
	Volume of hemisphere = $\frac{2}{3} \times \pi \times 3^3$	**1**
	= 18π cm^3	½
	Volume of ice-cream = $30\pi + 18\pi$	½
	= 48π cm^3 (≈ 150.86 cm^3)	½
35. (A)	Mode = 55 ⇒ modal class is 45 – 60 l = 45, f_1 = 15, f_0 = x, f_2 = 10, h = 15	½
	$55 = 45 + \left(\frac{15 - x}{2 \times 15 - x - 10}\right) \times 15$	**1**
	⇒ 2(20 – x) = 3(15 – x)	½
	⇒ x = 5	½
	<table><tr><th>C.I.</th><th>f_i</th><th>x_i</th><th>$f_i x_i$</th></tr><tr><td>0 – 15</td><td>10</td><td>7.5</td><td>75</td></tr><tr><td>15 – 30</td><td>7</td><td>22.5</td><td>157.5</td></tr><tr><td>30 – 45</td><td>5</td><td>37.5</td><td>187.5</td></tr><tr><td>45 – 60</td><td>15</td><td>52.5</td><td>787.5</td></tr><tr><td>60 – 75</td><td>10</td><td>67.5</td><td>675</td></tr><tr><td>75 – 90</td><td>12</td><td>82.5</td><td>990</td></tr><tr><td>Total</td><td>59</td><td></td><td>2872.5</td></tr></table>	**Correct Table** **1½**
	Mean = $\frac{2872.5}{59}$	½
	= 48.69 (approx.)	½
	OR	

35. (B)

Height (in cm)	Number of girls (f)	cf
Below 140	4	4
140 – 145	7	11
145 – 150	18	29
150 – 155	11	40
155 – 160	6	46
160 – 165	5	51

Correct Table **1½**

$\frac{N}{2}$ = 25.5 ∴ Median class is 145 – 150

l = 145, cf = 11, f = 18, h = 5 **½**

Median = $145 + (\frac{25.5 - 11}{18}) \times 5$ **1**

= 149.03 cm **½**

3 Median = Mode + 2 Mean **½**

⇒ Mean = $\frac{3 \times 149.03 - 148.05}{2}$ **½**

= 149.52 cm **½**

Section – E

[This section comprises solution of 3 case-study/passage-based questions of 4 marks each]

Q. No.	HINTS/SOLUTION	Marks
36.	(i) Common differences are 3 and – 3 Sum of common differences = 3 + (– 3) = 0	**1**
	(ii) a_{34} = 187 + 33 × (– 3) = 88	**1**
	(iii) **(A)** $S_{10} = \frac{10}{2}$ [2 × (– 5) + 9 × 3]	**1**
	= 85	**1**
	OR	
	(iii) **(B)** Let n^{th} terms of the two progressions be equal. – 5 + (n – 1) × 3 = 187 + (n – 1) × (– 3)	**1**
	⇒ 6(n – 1) = 192 ⇒ n = 33 So, 33rd term of the two progressions will have the same value.	**1**
37.	(i) PR = $\sqrt{(8-2)^2 + (3-5)^2} = \sqrt{40} = 2\sqrt{10}$ units	**1**
	(ii) Mid-point of PR = $(\frac{2+8}{2}, \frac{5+3}{2}) = (5, 4)$ As (5, 4) ≠ Q (4, 4), Q is not the mid-point of PR.	**1**

	(iii) **(A)** Let the point on x-axis be A (x, 0). AP = AQ $\Rightarrow (x-2)^2 + 25 = (x-4)^2 + 16$ $\Rightarrow 4x = 3 \Rightarrow x = \frac{3}{4}$ Required point is $(\frac{3}{4}, 0)$ **OR** (iii) **(B)** Coordinates of S = $(\frac{2\times4+3\times2}{2+3}, \frac{2\times4+3\times5}{2+3})$ $= (\frac{14}{5}, \frac{23}{5})$	**1** **1** **1** **1**
38.	Height of India Gate above Shreya's eye level = 42 – 1 = 41 m (i) $\tan\theta = \frac{41}{41} = 1 \Rightarrow \theta = 45°$ (ii) $\tan 60° = \frac{41}{d} \Rightarrow d = \frac{41}{\sqrt{3}}$ m or $\frac{41\sqrt{3}}{3}$ m (iii) **(A)** At 45°: $\frac{41}{x} = \tan 45° \Rightarrow x = 41$ m At 30°: $\frac{41}{y} = \tan 30° = \frac{1}{\sqrt{3}} \Rightarrow y = 41\sqrt{3}$ m Distance moved back = $41\sqrt{3} - 41 = 41(\sqrt{3}-1)$ m **OR** (iii) **(B)** $\tan\theta = \frac{41}{\frac{41}{\sqrt{3}}} = \sqrt{3}$ $\Rightarrow \theta = 60°$	**1** **1** **1** **1** **1** **1**